

From Statistical Certification to Ledger Accounting

Q

= query



= RAG plans

α

= contract risk budget

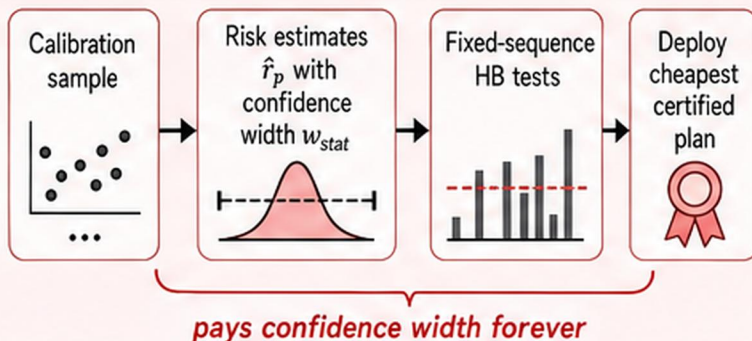


= decline (bot)

B_t

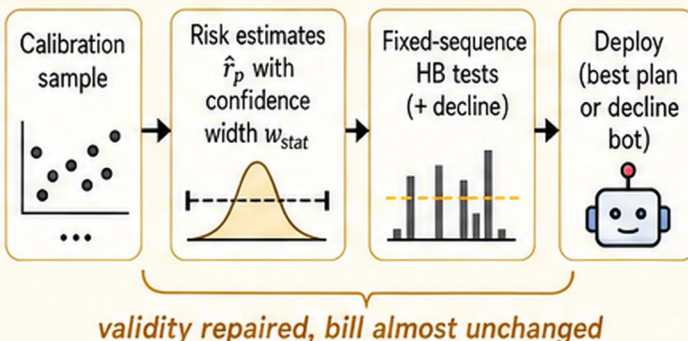
= ledger

(a) One-shot LTT Certification



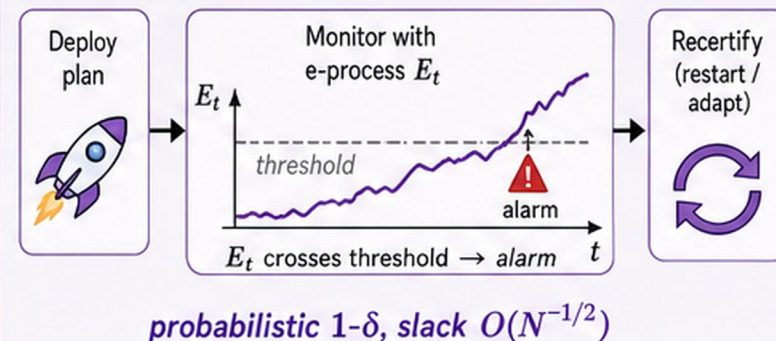
Pathwise $\times$ | Below risk floor $\times$ | $|P|$ -independent $\times$

(b) LTT + Decline



Pathwise $\times$ | Below floor $\checkmark$ | Cost still high $\times$

(c) Online e-process / CSA-style



Anytime-valid $\checkmark$ | Pathwise exact $\times$ | No δ $\times$

(d) Ours: Ledger-Gated Execution

Q
Query

We control cost while guaranteeing pathwise risk.



1 Plan menu P

Plans $P_1, \dots, P_K$ with costs and risks

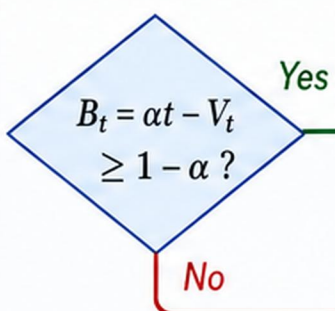
Plan	Cost c_p	Risk $\hat{r}_p$
P_1	c_1	$\hat{r}_1$
P_2	c_2	$\hat{r}_2$
$\vdots$	$\vdots$	$\vdots$
P_K	c_K	$\hat{r}_K$

Lower cost $\rightarrow$ prefer
Lower risk $\rightarrow$ safer



2 Ledger Gate

Test the budget condition



3 Act

Choose action



Sampled plan from Action LP mixture

Sample plan P_p with prob. x_p (cost-optimal under gate)



Decline bot

Return decline response (loss = 0)

4 Observe & Update

Account the outcome



Observe loss ℓ_t (0-1 loss for step t)



Update cumulative loss $V_t = V_{t-1} + \ell_t$



Update ledger $B_t = \alpha t - V_t$



$V_t \leq \alpha t$
on **EVERY** path —
no δ , no exchangeability,
no horizon.

Pathwise risk guarantee with ledger accounting.



Observe loss

See what happened



Account on ledger

Update V_t and B_t



Optimise cost only

Risk handled by the ledger gate